

Education

- **University of Wisconsin-Madison** Madison, WI
 - *4.0 GPA, Computer Science Major, Starting Grad School Fall 2027* *September 2024 - May 2027*
 - Taken: Deep Learning for NLP, Intro to Algorithms, Machine Organization and Programming, Intro to AI, Intro to Computer Engineering, Intro to Discrete Mathematics

Relevant Experience

- **Extreme Engineering Solutions (X-ES)** Verona, WI
 - *Embedded Software Engineering Co-op* *May 2026 - December 2026*
 - Authored Linux and VxWorks device trees for two new rugged embedded boards by cross-referencing hardware schematics, processor manuals, and device trees from similar existing boards.
 - Used those device trees to build and validate internal Buildroot-based Linux releases for both boards, integrating board-support configurations and testing extensively.
 - Produced customer-facing Gentoo Linux releases for both boards by debugging a legacy build pipeline, fixing broken API calls and migrating scripts from Python 2 to Python 3; authored an internal guide documenting the process for future engineers.
 - Additionally, validated a new release of X-ES Enterprise Linux (XEL), a RHEL-based OS distribution, by running the integration test suite and manually verifying command output and system logs against expected results, then updated the corresponding product manual.
- **Wisconsin Autonomous Vehicle Club** Madison, WI
 - *Perception Lead* *January 2025 - present*
 - Rewrote lane segmentation (drivable area detection) for our autonomous car's perception stack in C++ using TensorRT; combined with teammates' object detection and lane line rewrites, contributed to a 142% increase in framerate.
 - Leading a 10-20 member perception team building the vehicle's real-time computer vision pipeline using 3D stereo cameras.
 - The team placed 2nd of 10 in AutoDrive II for the past two years.
- **Undergraduate TA** Madison, WI
 - *CS540: Intro to AI* *Spring Semester 2026*
 - Held weekly office hours, helping students work through homework assignments and understand topics ranging from linear regression and unsupervised learning to transformers and reinforcement learning.
- **Internship with the Institute for Computing in Research** Austin, TX
 - *Intern* *July 2024*
 - Created a neural network to play chess under the direction of Los Alamos scientist Mark Galassi.
 - Code: <https://github.com/asventon1/chess2> Paper: <https://tinyurl.com/3spr2vk7>
- **Personal Projects**
 - *My own creations (with videos for some and code for all)* *2017 - present*
 - Google street view mesh generator: <https://github.com/asventon1/goondaMaps>
Generates a 3D point cloud from a Google Street View panorama.
 - Hamster tracker: <https://github.com/asventon1/oreo>
Makes use of an Arduino and proximity sensors to track and graph the activity of a hamster wheel.
 - HEB slot getter: <https://github.com/asventon1/slot-getter>
Checks the HEB grocery store website for curbside slots and notifies you if there is an open one.

Qualifications

- Languages: Python, C, C++, Java, JavaScript, Bash
- Embedded & Systems: Linux (Buildroot, Gentoo, RHEL), VxWorks, Device Trees, Serial Console, Arduino
- Tools: Git, Make, CMake, Docker
- ML/Computer Vision: NumPy, TensorFlow, PyTorch, TensorRT